

Full Project Blueprint — Telecom Coverage Clustering Project

Mapped against the full report index (Sections 1–6)

How to read this document

This is **not a product/service project** — it's a data analytics project (clustering + descriptive analytics on telecom tower data), and that's completely fine. The report index is a generic template used across all project types, so every section below has been reframed to genuinely fit an analytics project instead of forcing a fake "business/user" story.

Two tracks, side by side:

- **Part A — Build Roadmap:** the actual technical steps you and your teammate execute, in order. Tick these off as you finish real work.
- **Part B — Report Mapping:** once a Part A step is done, this tells you exactly which report section it feeds into, so writing the report becomes "describe what we already did," not a separate task from scratch.

On the "use case diagram" confusion specifically: your dashboard IS the interactive part of the system — so the person who opens the dashboard and clicks around IS your "user" for diagramming purposes. This doesn't require inventing a business, a client, or a fake stakeholder. One actor (call it "User" or "Analyst") interacting with a dashboard is a completely legitimate, honest use case diagram for exactly the project you're building.

PART A — Build Roadmap

Phase 1 — Data Preparation

- ☒ **Merge raw tower data:** combine 404.csv + 405.csv into one unified tower-level table (*marked complete per your update*)
- ☒ **Decode carriers:** join MCC-MNC India.csv lookup onto the combined table to convert MNC codes into carrier names (*marked complete — this is what you meant by "central dataset"*)
- ☐ Run basic data quality check: `.info()`, `.isnull().sum()`, confirm row count, confirm AverageSignal is indeed mostly unusable as expected
- ☐ Clean invalid/missing lat-long values (drop rows outside India's bounding box)
- ☐ Deduplicate if duplicate Cell ID/location entries exist

Phase 2 — Regional Enrichment (State level)

- ☐ Load `states_india.geojson`
- ☐ Spatial join: assign each tower a state, using geopandas
- ☐ Aggregate to state level: tower count, radio-type mix (%2G/3G/4G/5G), carrier count, Range average, Samples average, Changeable=0 ratio

Phase 3 — Regional Enrichment (District level)

- ☐ Source the external district boundary file (DataMeet/LGD)
- ☐ Spatial join: assign each tower a district
- ☐ Aggregate to district level — same feature set as Phase 2, at finer granularity

Phase 4 — Feature Engineering

- ☐ Finalize the feature set used for clustering (from Phases 2 & 3 aggregates)
- ☐ Scale features (StandardScaler) separately for the state-level and district-level tables
- ☐ Construct a rule-based coverage tier label (well-served/moderate/underserved) as a sanity-check reference for later

Phase 5 — Clustering

- ☐ Run K-Means on state-level data; determine K via elbow method + silhouette score
- ☐ Run K-Means on district-level data; determine K (may differ from state-level K)
- ☐ Run DBSCAN on both, as a comparison method
- ☐ Interpret and label each cluster (e.g., "Cluster 1: Dense urban 4G/5G")
- ☐ Cross-check clusters against your rule-based tier labels from Phase 4

Phase 6 — Optional Supervised Extension

- ☐ Train a classifier (e.g., Random Forest) to predict coverage tier/cluster from features
- ☐ Evaluate with accuracy/precision/recall/confusion matrix
- ☐ Run feature importance or SHAP to explain predictions

Phase 7 — Dashboard

- ☐ Build Streamlit app skeleton
 - ☐ State-level map view (colored by cluster)
 - ☐ District-level map view (colored by cluster), with toggle between the two
 - ☐ Click-to-inspect region detail panel
 - ☐ Carrier comparison charts
 - ☐ Radio-generation distribution charts
 - ☐ Cluster explanation panel
-

PART B — Report Section Mapping

1. ABOUT THE SYSTEM

1.1 Problem Definition (Identification of needs)

- ☐ Write once Phase 1 is fully done. Content should state: raw tower-level data exists in bulk but is unusable for regional decision-making as-is (too granular, no regional grouping, no coverage classification, one unreliable field). State the analytical need: a pipeline that aggregates and classifies regions by coverage quality.
- ☐ No external "client" needed — frame the need as: understanding India's regional mobile coverage patterns from open data, similar to how a telecom planning team would approach it.

1.2 Requirement Specifications

- ☐ Write once Phases 1–3 are scoped. Functional requirements = what your pipeline+dashboard must do (ingest, clean, aggregate at 2 granularities, cluster, visualize, allow region inspection). Non-functional = performance, usability, modularity, no paid dependencies.

1.3 Tools and Technology Used

- ☐ Write anytime — this is just your confirmed tool list: Python, pandas, geopandas, scikit-learn, (SHAP if Phase 6 done), Streamlit, Plotly/Folium, Jupyter. List the datasets too (Kaggle Mobile Network Coverage India + DataMeet district boundaries).

2. SYSTEM DESIGN USING UML

2.1 Data Flow Diagrams (Level 0 & 1)

- ☐ Write once Phases 1–5 are structurally finalized (doesn't need to be fully coded, just the stages decided).
- ☐ **Level 0:** User $\rightleftharpoons$ System (single bubble) $\rightleftharpoons$ Data stores (Raw Tower Data, Cluster Output)
- ☐ **Level 1:** expand into your real pipeline stages as separate process bubbles — this should mirror Part A's phases almost exactly:
 - Process 1: Data Cleaning & Carrier Decoding (Phase 1)
 - Process 2: Regional Aggregation — State (Phase 2)
 - Process 3: Regional Aggregation — District (Phase 3)
 - Process 4: Clustering (Phase 5)
 - Process 5: Dashboard Rendering (Phase 7)
 - Data stores between each: Cleaned Data $\rightarrow$ State-Aggregated Data / District-Aggregated Data $\rightarrow$ Cluster-Labeled Data
- ☐ This DFD is essentially "Part A, drawn as a diagram" — once Part A's structure is locked, this section is mostly transcription, not new thinking.

2.2 Use Case Diagram

- ☐ Write once Phase 7 (dashboard) scope is decided (doesn't need to be built yet, just planned).
- ☐ **Actor:** User (single actor is enough and honest for this project — no need to invent more)
- ☐ **Use cases**, drawn from your actual planned dashboard features (Phase 7):
 - View State-Level Coverage Map
 - View District-Level Coverage Map
 - Select a Region for Details
 - View Carrier Comparison Charts
 - View Radio-Generation Distribution
 - View Cluster Explanation
- ☐ Keep it to what you're actually building — 6 use cases, 1 actor is a clean, honest, appropriately-scoped diagram for this project. Don't pad it.

2.3 Class Diagram (*beyond current submission — for later*)

- ☐ When you get here: model your actual code structure as

classes — e.g., `DataLoader`, `RegionAggregator`, `ClusterModel`, `DashboardApp` — reflecting real modules in your codebase, not an invented object-oriented business domain.

2.4 Sequence Diagrams (*beyond current submission — for later*)

- ☐ When you get here: show the sequence of calls for one key interaction, e.g., "User selects District view" → `DashboardApp` requests data → `ClusterModel` returns labels → Map renders.

3. DATA DICTIONARY (*beyond current submission — for later*)

3.1 Database tables

- ☐ When you get here: document your final aggregated tables' schema (column names, types, meaning) — this is literally the column table you already have from the dataset inspection work done earlier, extended with your engineered features.

4. IMPLEMENTATION (*beyond current submission — for later*)

4.1 Screen Layouts with validations

- ☐ Screenshots of your actual Streamlit dashboard screens, with notes on any input validation (e.g., handling a region with no data gracefully).

4.2 Sample Coding (with Security tasks)

- ☐ Include key code snippets (cleaning function, clustering call, dashboard callback). "Security tasks" here can reasonably mean input validation/error handling in your code, since there's no user login/auth in this project — state that explicitly rather than forcing a fake login system.

4.3 Steps of execution and Navigation

- ☐ How to run the project end-to-end (run pipeline script → launch Streamlit app → navigate the dashboard).

5. CONCLUSION (*beyond current submission — for later*)

5.1 Importance of the worked-out tasks

- ☐ What the clustering reveals and why it matters (coverage gap identification).

5.2 Further Enhancement in future

- ☐ E.g., live data refresh, adding population data, extending to a supervised prediction system.

6. REFERENCES (*beyond current submission — for later*)

6.1 Books, Research Papers, Datasets, Tools

- ☐ Cite: OpenCellID, Kaggle dataset page, DataMeet/LGD district boundaries, scikit-learn/Streamlit documentation, any clustering methodology papers you reference for justifying K-Means/DBSCAN choice.
-

Where you stand right now

Based on what you've told me: **Phase 1 is complete.** Next up is Phase 2 (state-level regional enrichment) — that's your immediate next technical step, and once it's done you'll have enough to start drafting Section 1.1–1.3 of the report in parallel with starting Phase 3 (district-level).